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Proceedings of the Combustion Institute 33 (2011) 185–192
Proceedings of the Combustion Institute
www.elsevier.com/locate/proci

Detailed chemical kinetic reaction mechanisms for primary reference fuels for diesel cetane number and spark-ignition octane number

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Available online 24 August 2010

Abstract

A detailed chemical kinetic reaction mechanism is developed for primary reference fuel mixtures of *n*-hexadecane and 2,2,4,4,6,8,8-heptamethyl nonane for diesel cetane ratings. The mechanisms are constructed using existing rules for reaction pathways and rate expressions developed previously for the primary reference fuels for gasoline octane ratings, *n*-heptane and *iso*-octane. These reaction mechanisms are validated by comparisons between computed and experimental results for shock tube ignition and for oxidation under jet-stirred reactor conditions. The combined kinetic reaction mechanism contains the submechanisms for the primary reference fuels for diesel cetane ratings and submechanisms for the primary reference fuels for gasoline octane ratings, all in one integrated large kinetic reaction mechanism. Representative applications of this mechanism to several test problems are presented, describing fuel/air autoignition variations with changes in fuel cetane and octane numbers, and others describing fuel combustion in a jet-stirred reactor environment with the fuel varying from pure 2,2,4,4,6,8,8-heptamethyl nonane (cetane number of 15) to pure *n*-hexadecane (cetane number of 100).
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Keywords: Mechanisms; Oxidation; Kinetic mechanisms

1. Introduction

Performance of reciprocating engines under normal operating conditions depends rather sensitively on fuel ignition properties. Diesel fuel ignition is rated by its cetane number (CN), where fuels with higher values of CN ignite more readily than fuels with lower CN values. Gasolines are rated by their octane number (ON), where higher values of ON indicate greater resistance to autoignition and therefore lower potential to experience engine knock. Each of these ignitability scales is defined in terms of a pair of specific primary reference fuels (PRF) with ignition properties as widely different as possible. PRF components assigned to diesel fuel are both C_{16} molecules, while PRF components for gasoline are a C_7 and a C_8 hydrocarbon, reflecting the general fuel size distributions for both fuels. Both scales are similar in that each has a very easily ignited PRF *n*-alkane molecule, *n*-heptane (n-C_7H_{16}) for gasoline and *n*-hexadecane (n-$C_{16}H_{34}$, *n*-cetane) for diesel fuel, and each has a

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doi:10.1016/j.proci.2010.05.087

highly branched *iso*-alkane component that is difficult to ignite, *iso*-octane (2,2,4-trimethyl pentane, i-C_8H_{18}) for gasoline and *iso*-cetane (2,2,4,4,6,8,8-heptamethyl nonane, HMN, i-$C_{16}H_{34}$) for diesel fuel. By definition, *iso*-octane has an ON of 100, while *n*-heptane has an ON equal to 0. Also by definition, *n*-hexadecane has a CN equal to 100 while *iso*-cetane has a CN of 15. Structures of all four of these PRF fuels are shown in Fig. 1.

We previously developed detailed comprehensive chemical kinetic reaction mechanisms for the gasoline PRF components *n*-heptane [1] and *iso*-octane [2], including both high temperature and low temperature submechanisms. Development of these reaction mechanisms was facilitated by defining 25 classes of reactions [1,2] so that a series of reaction rate rules could be employed in a systematic way. Detailed chemical kinetic reaction mechanisms using the same reaction pathway and rate classes have provided kinetic tools for studies of many other fuels [3–11], examining the role of fuel molecular structure on hydrocarbon combustion, and they also provided a starting point to development of more general surrogate fuels for gasoline [12] and diesel fuel [13].

We have now developed new individual kinetic reaction mechanisms for *n*-hexadecane [11] and for *iso*-cetane [14], using the same reaction classes as those noted above [1,2], and each was separately validated by comparison with the very few experimental studies that were available. In the present study, the individual mechanisms for each diesel reference fuel are combined into a single, unified mechanism that can be used to simulate and interpret the effects of mixing the components in diesel engine and other studies. The companion combined gasoline PRF mixture mechanism has been available for some years, so now a similar kinetic modeling capability exists for both major engine environments and is widely available to those who want to use it in numerical studies [15]. Below, we use the combined PRF mechanisms to examine several general problems in PRF kinetics that illustrate important features of reference fuels.

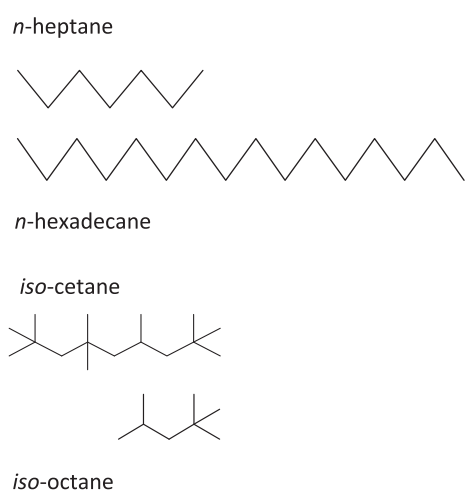

Fig. 1. Primary reference fuels for gasoline and diesel fuel.

2. Kinetic features of individual PRF components

Validation studies of the present *n*-cetane kinetic mechanism have previously been reported [11] using experimental result of Ristori et al. [16] in a jet-stirred reactor which reported *n*-cetane oxidation at high temperatures above 900 K. Ristori et al. also included kinetic modeling results for these *n*-cetane oxidation experiments, and Dagaut et al. [17] later used this same high temperature *n*-cetane kinetic model to simulate oxidation of rapeseed methyl ester (RME), a mixed methyl ester with carbon chains as long as 17C atoms. For the most part, the *n*-cetane mechanism was a very successful surrogate for the RME fuel. More recently, Biet et al. [18] and Westbrook et al. [11] developed detailed kinetic mechanisms for *n*-alkanes up to *n*-hexadecane that both included high temperature and more complex low temperature reaction pathways.

The Biet et al. study included new experimental results for oxidation of mixtures of *n*-decane and *n*-hexadecane under PSR conditions, and we have used these experimental results to further validate our *n*-cetane reaction mechanism. Biet et al. examined the oxidation of a stoichiometric, atmospheric pressure mixture of 1480 ppm *n*-decane and 520 ppm *n*-hexadecane, diluted in helium, at temperatures from 550 to 1100 K, with a residence time of 1.5 s, which provides particular emphasis on the region of negative temperature coefficient (NTC). The present kinetic mechanism was used to simulate these experimental results, with overall very good agreement. The key feature of the experiments was pronounced NTC behavior at temperatures from 650 K to nearly 800 K, shown by the fuels and CO profiles in Fig. 2, and the corresponding computed results show the same NTC behavior. Similar computed results and agreement with experimental results were found for most important species in this study. The degree of agreement using the present kinetic mechanism is very close to that also reported by Biet et al. in their own kinetic modeling analysis [18].

Dagaut and Hadj-Ali [19] studied *iso*-cetane combustion, both experimentally and computationally, in a jet-stirred reactor. Their experiments covered a temperature range from 770 to 1070 K and used a high temperature kinetic mechanism from Agosta et al. [20] and another detailed mechanism from Battin-Leclerc et al. [21] to analyze their experimental results. Subsequent kinetic modeling by Ranzi et al. [22] refined the mechanisms and added a lumped approach for the low temper-

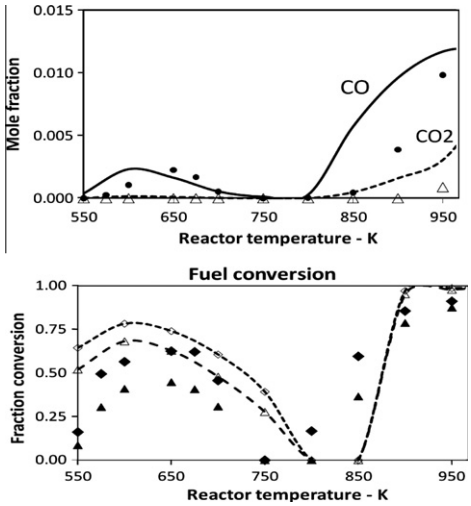

Fig. 2. Experimental results from Biet et al. [18] (symbols) and computed (lines) species concentrations in PSR oxidation of stoichiometric mixtures with 1480 ppm *n*-decane and 520 ppm *n*-cetane in helium, at atmospheric pressure and a residence time of 1.5 s. Upper frame shows CO (solid line and circles) and CO_2 (dashed line and triangles). Lower frame shows. *n*-decane (short dashed line and diamonds) and *n*-hexadecane (long dashed line and triangles).

ature kinetic regime, thoroughly describing the NTC behavior seen at temperatures below 750 K. In another high temperature study, Oehlschlaeger et al. [14] carried out shock tube ignition delay experiments and used a detailed kinetic mechanism to analyze their results, with very good agreement between experiments and kinetic modeling results.

The structure of *iso*-cetane shown in Fig. 1 is closely related to the structure of *iso*-octane. Breaking the central C—C bond in *iso*-cetane produces two C_8H_{17} octyl radicals, both of which are also produced in *iso*-octane oxidation. Thus the ignition of *iso*-cetane is very similar to *iso*-octane, especially the high level of *iso*-butene that is the primary intermediate in the combustion of both fuels. However, by coupling the two *iso*-octyl radicals together, the resulting *iso*-cetane molecule has 6 secondary C—H bonds while *iso*-octane has only 2; both fuels have one tertiary C—H bond. The existence of multiple secondary C—H sites in *iso*-cetane makes it much easier for intramolecular RO_2 isomerization reactions to occur at significant rates, while there are no comparable reaction pathways in *iso*-octane [3]. These factors combine to make *iso*-cetane more reactive than *iso*-octane in the low temperature regime, although since *iso*-cetane has such a high fraction of primary C—H bonds, it is still considerably less reactive than *n*-alkanes at low temperatures. As shown below, *iso*-cetane autoignition falls between the ignition rates of *iso*-octane and the *n*-alkanes.

The mechanism for *n*-heptane is a subset of the mechanism for *n*-cetane, and the mechanism for *iso*-octane is a subset of that for *iso*-cetane, so the present kinetic mechanism for diesel PRF mixtures includes kinetic reaction submechanisms for the ON PRFs, and the entire mechanism thus includes the PRF mechanisms for both gasoline and diesel fuels. The resulting diesel PRF mechanism, combining both *n*-cetane and *iso*-cetane, is quite large, with more than 2800 chemical species and nearly 11,000 elementary reactions, although typical homogeneous ignition calculations of autoignition of these diesel PRF mixtures can be done in 5 min of CPU time on a laptop computer. Mechanism reduction techniques will be required to use these mechanisms in CFD applications.

A wide variety of mechanism validations for *n*-hexadecane were included in our previous study of *n*-alkane fuel mechanisms [11]. Perhaps most interesting, that study indicated that the ignition rates of all of the *n*-alkanes from *n*-C_7H_{16} to *n*-$C_{16}H_{34}$ are almost identical, at least for stoichiometric mixtures of each *n*-alkane in air at pressures of 13.5 and 40 bar pressure. This is illustrated in Fig. 3, showing experimental shock tube ignition delay results from Ciezki et al. [23] at 13.5 bar pressure for stoichiometric mixtures of *n*-heptane and air as filled triangles. Similar experiments using *n*-decane as the fuel [24] produced ignition delay times very close to the *n*-heptane values in Fig. 3. Computed ignition delay times for *n*-heptane and *n*-cetane are shown in Fig. 3, showing the very close agreement between them and with the experimental results. Recent experimental shock tube results of Shen et al. [25], using *n*-heptane, *n*-decane, *n*-dodecane and *n*-tetradecane strongly support the model prediction that the ignition rates of all these *n*-alkanes are very similar.

A recent study of shock tube ignition of *iso*-cetane, at the same conditions as those shown in Fig. 3, by Oehlschlaeger et al. [14] included kinetic

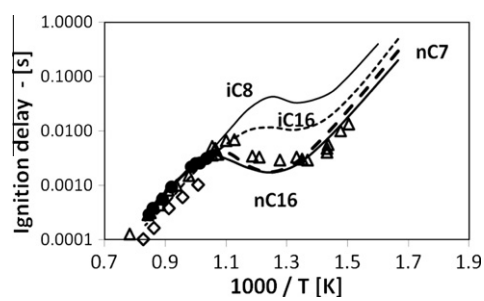

Fig. 3. Computed (lines) and experimental (symbols) ignition delay times for stoichiometric primary reference fuels in air at 13.5 bar. Light solid curve is *iso*-octane, short dashed curve is *iso*-cetane, long dashed curve is *n*-heptane, heavy solid curve is *n*-hexadecane. Triangles are *n*-heptane from Ciezki et al. [23], diamonds are *iso*-cetane [14], solid circles are *iso*-octane [26].

analysis using the present mechanism, and these experimental (filled diamonds) and kinetic modeling results are included in Fig. 3, showing excellent agreement. However, those experiments were limited to a temperature range from 879 to 1347 K and no low temperature phenomena were observed. To illustrate the influence of low temperature kinetic reaction pathways on *iso*-cetane ignition, the model at the same values of pressure and composition is extended to lower temperatures in Fig. 3, showing a modest negative temperature coefficient behavior over temperatures from about 700 to 800 K. Also shown for comparison is a curve showing comparable computational results for *iso*-octane. Experimental results for the same conditions for *iso*-octane by Fieweger et al. [26] are shown as filled circles in the figure and agree very well with the mechanism results but these also are available only for the higher temperature regime. It is clear from Fig. 3 that all of these *n*-alkane and branched alkane fuels have very similar ignition delays for high temperatures, above about 900 K, and quite similar ignition delays at temperatures below about 700 K, but they differ considerably within the NTC temperature range. The entire class of *n*-alkane fuels have extremely similar ignition delays for all temperatures, and within the NTC region *iso*-cetane ignites more slowly than the *n*-alkanes but faster than *iso*-octane. Similar computational [7] and experimental [27] studies of shock tube ignition of heptane isomers, whose octane numbers range from 0 to 112, are consistent with the results shown in Fig. 3, where the high temperature ignition delays are all very nearly equal but differ over the NTC region, with isomers having higher ON values igniting slower than isomers having lower ON values. We interpret these results to suggest that all of the gas phase, homogeneous kinetic factors influencing differences in octane and cetane ratings of saturated hydrocarbon fuels must be attributed to differences in their kinetic behavior within this rather narrow temperature range. Further kinetic analysis is needed to see if we can extend this conclusion to other classes of hydrocarbon fuels such as those with aromatic rings and double bonds, but recent studies of alkyl ester fuels such as those in biodiesel fuels [10] show very similar ignition behavior as that observed for the *n*-alkanes shown in Fig. 3.

We used recent experimental results from Dagaut et al. [16,19] from a jet-stirred reactor, with both *n*-hexadecane and *iso*-cetane (heptamethyl nonane) as fuels, to provide further validation of the kinetic reaction mechanisms. Selected results for *iso*-cetane are shown in Fig. 4, comparing representative experimental [19] and computed species concentrations for stoichiometric mixtures of *iso*-cetane and oxygen at 10 atm, diluted by nitrogen. The major species CO and CO_2 show excellent agreement with experiments, and other major intermediate species formaldehyde, *iso*-butene and propene also show very good agreement. Many other species also show very good agreement, while a few such as 1,3-butadiene are underpredicted by factors of as much as 5.

While there are a considerable number of experimental results for mechanism validation of the gasoline primary reference fuels *n*-heptane and *iso*-octane, there are limited results suitable for validating kinetic mechanisms for *n*-cetane and *iso*-cetane. Future improvements in this *iso*-cetane mechanism will require new experimental results to provide data for mechanism validation studies.

3. Primary reference fuel mechanism calculations

In this section, we use the combined PRF mechanisms to illustrate the capabilities of this mechanism. In Figs. 3 and 4, we used experimental results to validate the mechanisms for individual fuels, and here we use the coupled PRF fuels to illustrate a number of interesting features of both mixtures of fuel components.

The PRF mechanism for octane ratings (*n*-heptane and *iso*-octane) is imbedded within the PRF mechanism for cetane ratings, as noted above, and this mechanism was used to examine the ignition delay times for mixtures of *n*-heptane and *iso*-

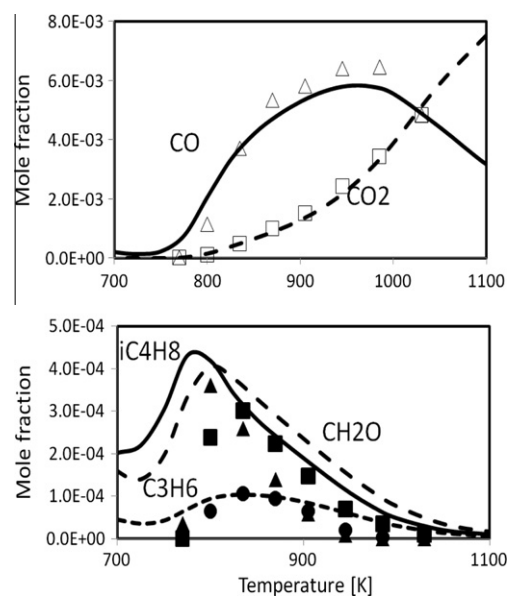

Fig. 4. Computed (lines) and experimental [17] (symbols) for stoichiometric *iso*-cetane oxidation in jet-stirred reactor at 10 atm, residence time of 1 s, 700 ppm fuel. In the bottom frame, the solid line and triangles show results for *iso*-butene, long dashed line and squares show formaldehyde, short dashed line and circles show propene. See Fig. 7 for fuel and hydrogen peroxide.

octane and illustrate some of the properties of these fuel mixtures. Computations were carried out for stoichiometric *n*-heptane/*iso*-octane/air mixtures with ON values of 0, 60, 80 and 100, all at 40 bar pressure. These results are compared with experimental measurements for the same conditions and PRF mixtures by Fieweger et al. [26] in Fig. 5. Experimental results are shown as symbols and the computed results as lines. The computed and experimental results agree extremely well for all of the fuel mixtures, adding confidence to the kinetic mechanisms for both of the ON reference fuels. All of the fuels show some NTC behavior, and the curves shift towards longer ignition delay times as the ON increases. At the same time, the local minimum in the ignition delay time moves to lower temperatures at ON increases. At 40 bar pressure, the NTC region extends to about 1000 K, while it extends only to about 900 K for the same fuels at 13.5 bar, as shown in Fig. 3. Similar to the observations from Fig. 3, the different fuels have different ignition delay times only for temperatures within a narrow temperature range between 700 and 950 K.

In Fig. 6, we repeat the same type of autoignition calculation for a range of stoichiometric diesel PRF mixtures at values of CN of 15, 50, 75 and 100. Similar to the computed results for ON in Fig. 5, the ignition delay times for the selected CN/PRF mixtures show a steady decrease in ignition delay as the CN is increased from 15 to 100, as well as the same phenomenon that the differences in ignition delay time seem to be negligible for all initial temperatures outside the range from 700 to 900 K. There appears to be a small difference at the lowest temperatures between the ignition delay times for *n*-cetane and the other CN/

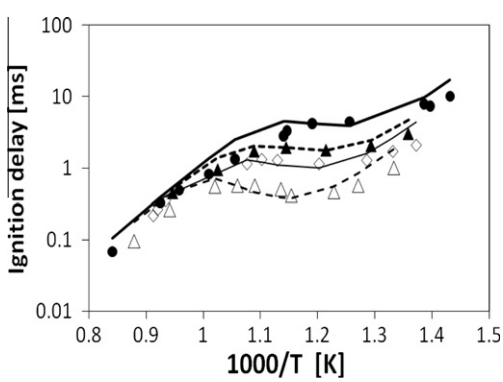

Fig. 5. Experimental (symbols) and computed (lines) values for ignition delay time for stoichiometric PRF/air mixtures at selected ON values. Data from [26] at 40 bar pressure, circles for ON = 100, squares for ON = 80, diamonds for ON = 60, asterisk for ON = 0. Heavy solid line is calculated values for ON = 100, heavy dashed line is ON = 80, light solid line is ON = 60, light dashed line is ON = 0.

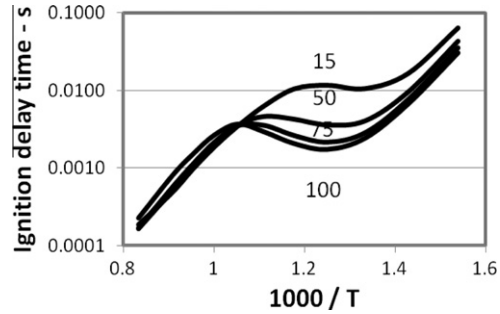

Fig. 6. Computed ignition delay times for stoichiometric diesel PRF/air mixtures at 13.5 bar pressure. Curves are labeled for CN = 15 (*iso*-cetane), 50, 75 and 100 (*n*-cetane).

PRF mixtures, but this difference is much less than the differences in the 700–900 K range. One cannot forget that cetane number tests, like actual diesel ignition in engines, depend on many factors in addition to homogeneous gas phase kinetics of ignition. Liquid fuel jet penetration and vaporization, mixing with residual gases and fresh air and heating of the combustion fuel/air mixture by the hot residuals and by piston motion, are also significant factors in diesel ignition. Since most of these other processes do not vary a great deal as the fuel composition changes over the range of conventional diesel fuels, the variations in kinetic properties shown in Fig. 6 certainly play a major role in diesel ignition and variations in CN.

Computed results for CN = 15 (i.e., pure *iso*-cetane) oxidation under PSR conditions were discussed above and compared with experimental results from Dagaut and Hadj-Ali [19] in Fig. 4. While those experiments did not show low temperature kinetic features, we used the kinetic model to extend the PSR results to 500 K, and the computed *iso*-cetane fuel profile and some of the intermediate species concentrations are labeled as "15" in Fig. 7. These results show a very pronounced NTC region from 600 to 700 K that was not observed in the experiments, which did not extend below 700 K but was shown in modeling analysis by Ranzi et al. [22]. We then repeated the same PSR simulations for PRF/CN mixtures of CN = 50, 75 and 100 (pure *n*-cetane), and some of these results are also shown in Fig. 7.

The simulations show that a considerable amount of reactivity takes place at temperatures between 550 and 700 K, and these results have significant impacts on distinctions related to variations in cetane numbers. All of these mixtures produce significant amounts of NTC behavior in the numerical JSR simulations, resulting in two major regions of considerable reactivity, first at about 600 K and the second beginning above 725 K. This can be seen in the results for fuel concentrations, showing that, for these experimental

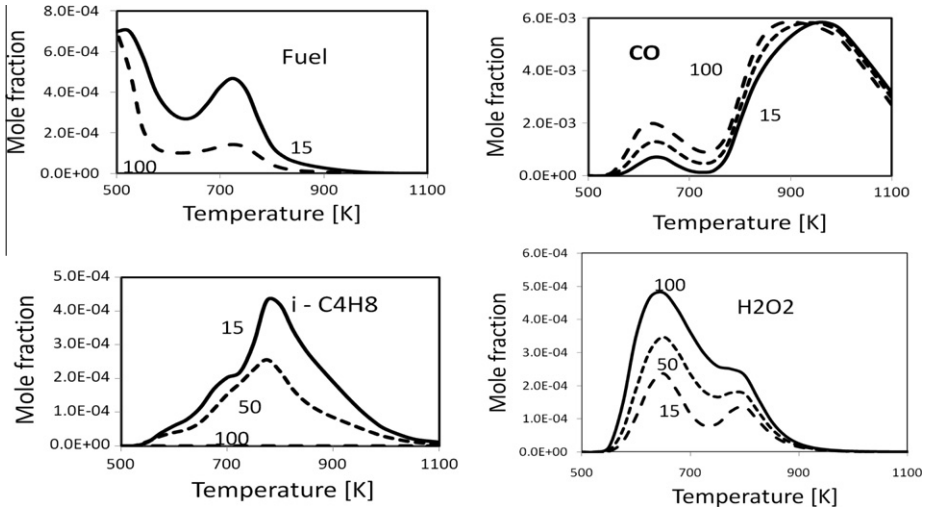

Fig. 7. Computed fuel and intermediate species concentrations under the same PSR conditions as in Fig. 4 for diesel PRF mixtures of CN = 15, 50 and 100. Fuel concentrations are 700 ppm, diluent is nitrogen, pressure is 10 bar, and the residence time is 1 s. CN number of each fuel mixture indicated on each frame.

conditions, 85% of the *n*-cetane is consumed at 600 K, while about 60% of the fuel is consumed when the fuel is *iso*-cetane. In Fig. 7, computed CO concentrations are overall similar for the three fuel mixtures, with significant relative differences seen only in the low temperature reaction zone at about 600 K. A significant trend can be seen in the computed profiles for *iso*-butene, which are highest in the case where the fuel is *iso*-cetane, and are extremely small in the case where the fuel is *n*-cetane. This is a result of the fact that *iso*-cetane, which reacts very much like *iso*-octane, reacts slowly specifically because it produces large amounts of *iso*-butene, which is quite stable. Other low CN fuels might not produce significant amounts of *iso*-butene but ignite slowly because they produce other stable intermediates. One implication of these "computational experiments" is that it would be very interesting to extend the actual experiments of Dagaut and Hadj-Ali, on which the computations are based, down to 550 K to fully explore the low temperature kinetic regime.

One species concentration that varied most with CN is hydrogen peroxide (H_2O_2) shown in Fig. 7, with high CN fuels producing considerably more H_2O_2 than lower CN fuels. Largest production of H_2O_2 is seen in the low temperature region between 600 and 700 K. In an engine, H_2O_2 produced at low temperatures decomposes at higher temperatures [28], producing two OH radicals which promote ignition. Thus production of high levels of H_2O_2 leads to high fuel reactivity and high cetane numbers. In a kinetic sense, production of large levels of *iso*-butene from *iso*-cetane slows production of H_2O_2 relative to mixtures with high initial levels of *n*-cetane, thus delaying ignition.

4. Discussion

The present work describes a new reaction mechanism for the primary reference fuels for both gasoline (octane ratings) and diesel fuel (cetane ratings), in a single model. This development has been made possible by growth in computer simulation capabilities and was carried out using established kinetic principles that apply existing rules for reaction pathways and rates for numerous classes of elementary reactions.

The present reaction mechanisms for all four fuels of both PRF mixtures were produced using the same reaction rate rules and approach described in detail by Curran et al. [1,2] for *n*-heptane and *iso*-octane. While some of the reaction classes and rate rules are somewhat simplified and improvements and additional detail could be made, the same level of approximation has been extremely productive in the cases of *n*-heptane and *iso*-octane, and we have chosen to retain the same level of approximation in the present mechanisms. Examples of improvements that will be included in the future include the results of recent work of Taatjes et al. [29] and others on alternative reaction pathways for reactions of alkyl radicals and O_2 to directly produce olefins via a concerted molecular elimination pathway, as well as site-specific reactions of olefins following recent studies of Mehl et al. [30] and VanHove et al. [31]. The ability of these mechanisms to describe kinetics of these fuels, with known limitations in the level of detail in the individual reaction pathways, deserves additional comment. Simplified submechanisms for the large olefin species are acceptable when the fuels being consumed are saturated

paraffin species, and rather small fractions of large olefins are produced only as intermediate species. An approximate submechanism for consumption of these minor intermediate components is adequate and works because it leads to most of the same smaller species that are produced by a more detailed olefin oxidation mechanism [30,31]. However, when the primary fuel includes components with double bonds, a more detailed olefinic species submodel is essential, and we have added such detailed submodels in such cases [32–34].

The success of the present low temperature, alkylperoxy radical submechanisms in the present models is more complex. The work of Taatjes et al. [29] and others shows that some reaction pathways in this region have been overlooked in the present model, yet the overall NTC region kinetic behavior of all of the present and other fuels is accurately described. The explanation of this lies in understanding the nature of the key features that must be described in this kinetic region. Each reaction pathway in the alkylperoxy radical isomerization regime provides a different overall combination of chain branching, chain propagation and chain termination. For example, the RO_2 isomerization pathway that proceeds via a 5-membered transition state ring and produces primarily HO_2 and an olefin species is a chain propagation pathway, producing one radical (i.e., HO_2) to replace the radical that produced the original R species. In contrast, the RO_2 isomerization pathway that proceeds via a 6-membered transition state ring is responsible for most of the chain branching in the NTC region, where subsequent reaction steps proceed through a second O_2 addition to the hydroperoxy alkyl radical and produce at least 2 OH and other radicals. These multiple radicals are essential to the low temperature ignition that is the first stage in 2-stage ignition of low ON hydrocarbons. It is the balance between such chain branching and propagation reaction pathways that is essential to describing low temperature ignition and then NTC behavior. When the new low temperature reaction pathways described by Taatjes et al. are added to the mechanism, they increase the fraction of low temperature chain propagation, and in order to reproduce observed NTC behavior, contributions from other, already existing, low temperature chain propagation reaction pathways must be reduced to compensate. This process of correcting the ratios of low temperature chain branching to chain propagation was used by Silke et al. [34] to reproduce NTC behavior in the ignition of cyclohexane when the direct molecular elimination reaction pathways described by Taatjes et al. were included. The present reaction mechanisms should be considered as works that are in continuous improvement, and these and other refinements are planned. At the same time, the success of the present mechanisms provides interesting insights about the essential features of such mechanisms, such as this example of the importance of the balance between chain branching and chain propagation pathways.

Further refinement of the diesel PRF portion of the model will become possible as more experimental validation data become available. The PRF model provides a computational tool to apply to a wide variety of combustion chemistry problems related primarily to diesel fuels, including a capability to suggest experiments that might be especially valuable for understanding important kinetic phenomena related to fuel chemistry. They also are a basis for systematic mechanism reduction to provide a computationally efficient diesel combustion reaction mechanism for use in multidimensional CFD simulations.

Acknowledgments

This work was supported by the US Department of Energy, Office of Vehicle Technologies, and the authors thank program managers Gurpreet Singh and Kevin Stork for their support. This work performed under the auspices of the US Department of Energy by Lawrence Livermore National Laboratory under Contract DE-AC52-07NA27344.

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